

ARDUINO-BASED TOUCHPAD TIC-TAC-TOE



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Tic-tac-toe or knots-and-crosses has traditionally been a pencil-and-paper two-player game wherein the player who first succeeds in placing three identical marks in a row (either horizontal, vertical or diagonal) wins the game. Presented here is a digital implementation of the tic-tac-toe game using a 3×3 array of bi-colour LEDs for

display, a simple resistive touchpad input with 3×3 key matrix and an Arduino board with ATmega328 controller. The bi-colour LED used is in red and green colours.

Let's say the first player is assigned red colour and the second green colour. The two players take turns to select the input switches on the touchpad. Touching the input switch causes the corresponding LED in the LED matrix to light up with the colour of the

player. The ATmega microcontroller on the Arduino board simultaneously keeps checking for a win or game-over situation.

If you don't have anybody to play with, you can switch to the one-player mode and play against the microcontroller instead.

Fig. 1 shows the functional block diagram of Arduino-based touchpad tic-tac-toe and the connections between the three constituent modules—resistive touchpad, Arduino board and 3×3 LED matrix.

Hardware

Fig. 2 shows the circuit of touchpad tic-tac-toe. The ATmega microcontroller on the Arduino board acts as the brain of the circuit. The touchpad provides analogue inputs while the LED matrix displays digital outputs.

Arduino board. Arduino is an electronics platform meant for rapid prototyping. The hardware is Open Source, allowing hobbyists, engineer-

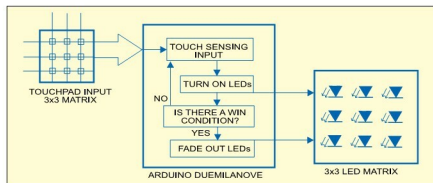


Fig. 1: Block diagram of Arduino-based touchpad tic-tac-toe

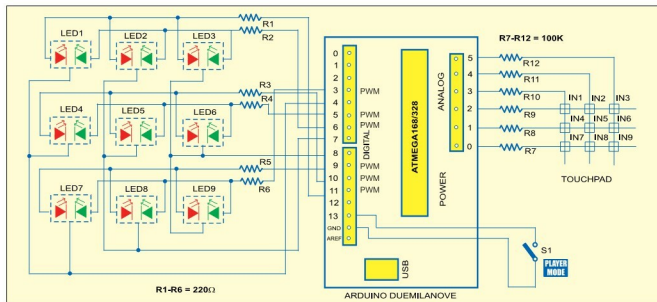


Fig. 2: Circuit of Arduino-based touchpad tic-tac-toe